

Hardware Efficiency & Worker Density

Hosting 100+ Concurrent Digital Employees on a Single Modest Server

Maximizing Hardware Density: The Economics of Efficiency

The true test of enterprise scalability is operational efficiency per server dollar. Because Fathom compiles to lean machine code with zero runtime interpreter overhead, it achieves unprecedented hardware worker density.

PYTHON FRAMEWORK SERVER FOOTPRINT (100 AGENTS)

- **RAM Required:** 64 GB – 128 GB RAM (Heavy Kubernetes Cluster).
- **CPU Saturation:** High context-switching drag and Python GIL contention.
- **Monthly Server Cost:** \$400 – \$1,200 / month on AWS/GCP.
- **Hosting Cost Per Worker:** \$4.00 – \$12.00 / month.

FATHOM RUST SERVER FOOTPRINT (100 COWORKERS)

- **RAM Required:** 1.5 GB – 3.5 GB RAM (Modest 8-Core Box).
- **CPU Utilization:** Zero idle CPU burn; epoll/kqueue event-driven I/O.
- **Monthly Server Cost:** \$30 – \$60 / month on Hetzner/DigitalOcean.
- **Hosting Cost Per Worker:** < \$0.50 / month.

Server Resource Allocation Model (8 Cores, 32GB RAM)

100+

CONCURRENT COWORKERS

~25 MB

AVERAGE RAM / COWORKER

< 1%

IDLE CPU UTILIZATION

99.4%

GROSS MARGIN ON COMPUTE

EDGE AND ON-PREMISE DEPLOYABILITY

Because Fathom's compiled binary requires only 15MB RAM, complete multi-agent swarms can be deployed directly on edge devices, local Mac Minis, or air-gapped industrial servers with zero cloud dependencies.

The Bottom Line: Fathom cuts server infrastructure costs by **95%**, allowing businesses to scale their virtual workforce profitably.